Heteropolysaccharides: Heteroglycans

- Polysaccharides: ≥ 10 monosaccharide units
- Heteropolysaccharides: ≥ 2 different monosaccharide units or their derivatives

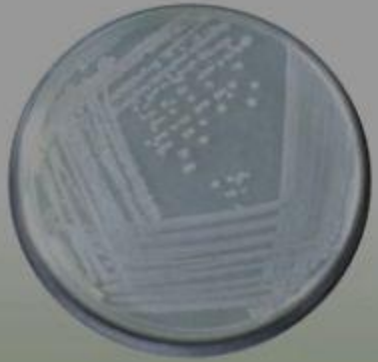
Agar, Agarose
Gum & Pectin
Hyaluronic acid
Chondroitin sulfate
Keratan Sulfate
Dermatan Sulfate
Heparan Sulfate

- Mucopolysaccharides: First discovered in Mucin (Mucous secretions)
- Glycosaminoglycans

Agar and Pectins

Agar
- Mostly found in seaweeds
- Polymer of galactose, sulfate and glucose
- Not digestible, Dietary fiber

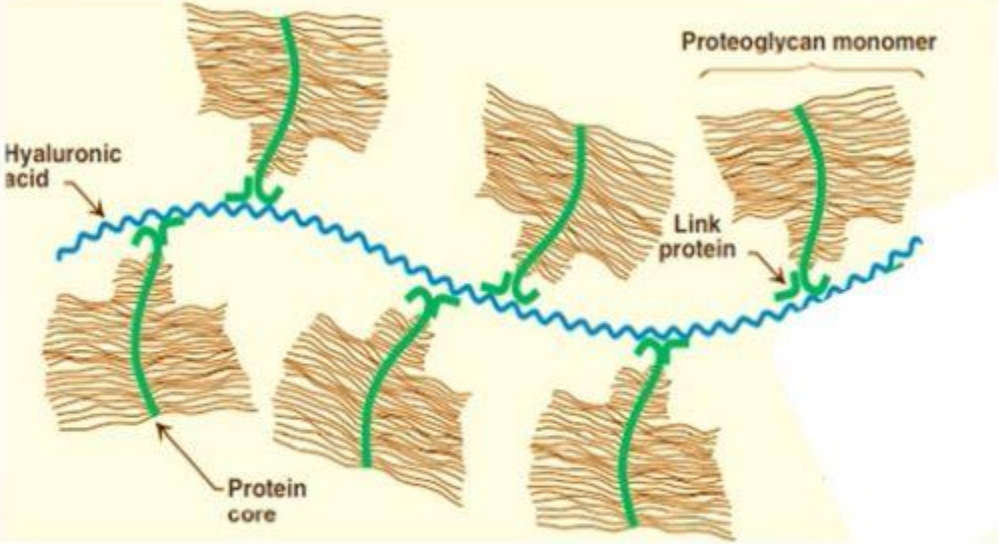
Agarose
- Galactose and anhydrogalactose
- Useful in laboratory as a major component of microbial culture media, and in electrophoresis



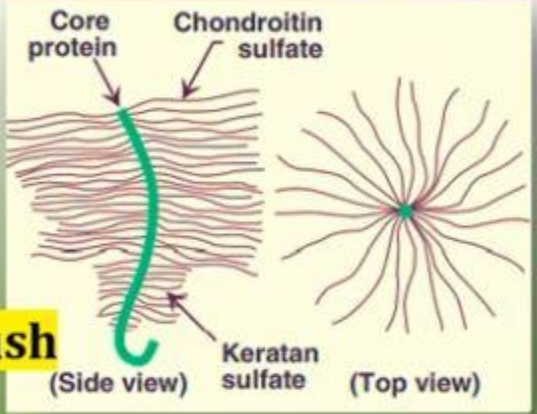
Pectins:
- Galactouronate and rhamnose
- Non digestible
- **Dietary fibers**
- Found in apples and citrous fruits
- Employed in **preparation of jellies**

Glycosaminoglycans: Definition/Properties/Functions

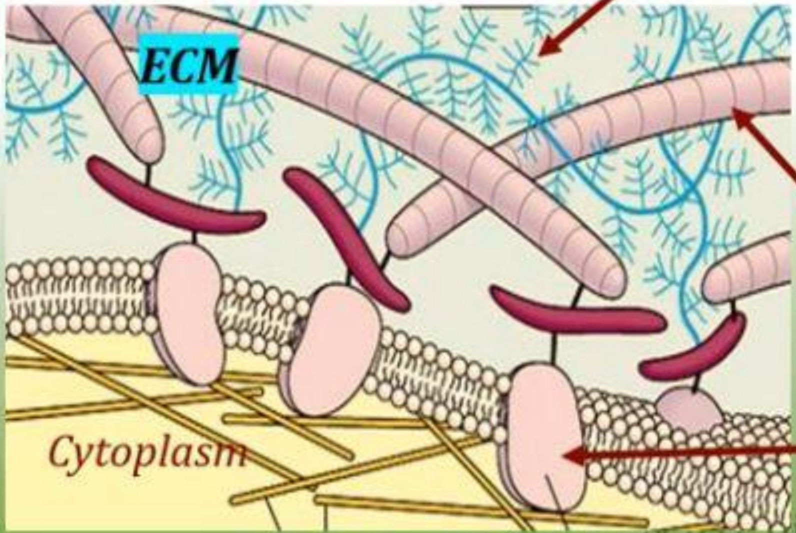
- Long,
- Negatively charged,
- Unbranched,
- Heteropoly saccharide chains
- Generally composed of a repeating disaccharide unit [acidic sugar–**amino sugar**]



Associated with a small amount of protein forming **proteoglycans.(>95% Carbohydrates)**



Bottlebrush



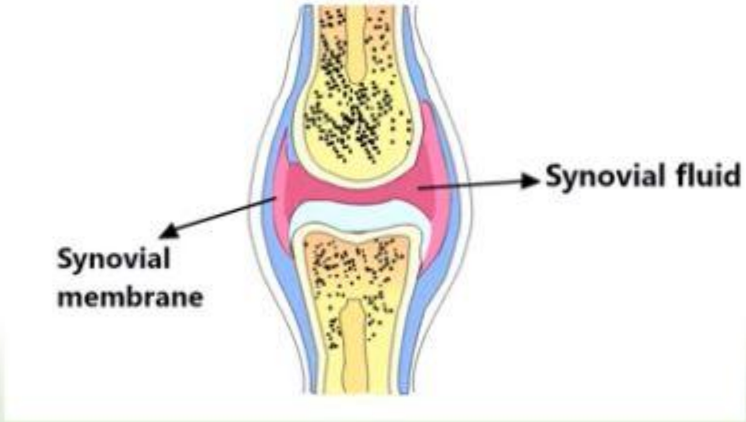
Proteoglycans aggregate

Collagen

Integrin

Glycosaminoglycans: Properties/Functions

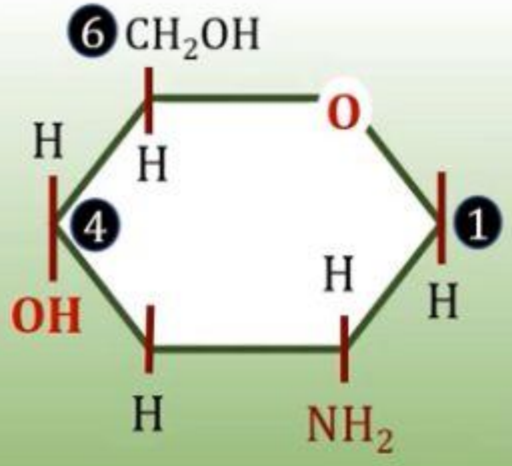
- Bind Large amount of water
- Viscous
- Lubricating



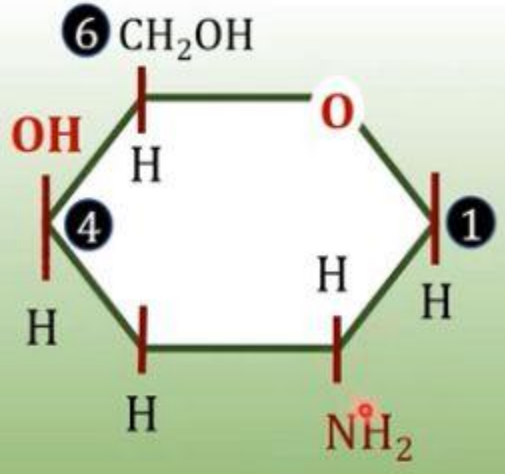
- Major component of Extracellular Matrix.
- Support cellular and fibrous components of tissues
- Mediate cell-cell signaling and adhesion

Glycosaminoglycans: Structure/Composition

- Amino sugars
  - Glucosamine
  - Galactosamine
- Uronic acids
  - Glucuronic acid
  - Iduronic acids



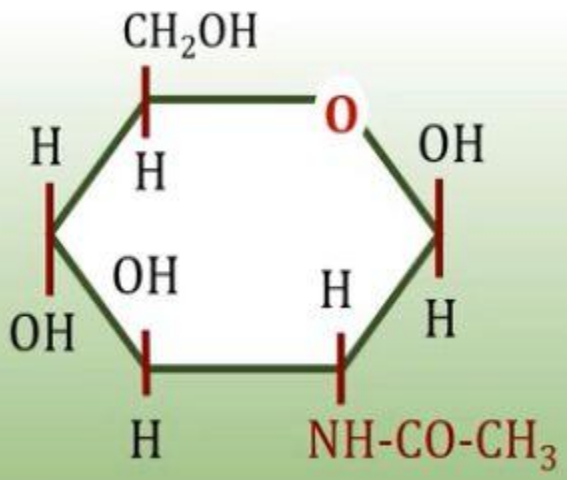
Glucosamine



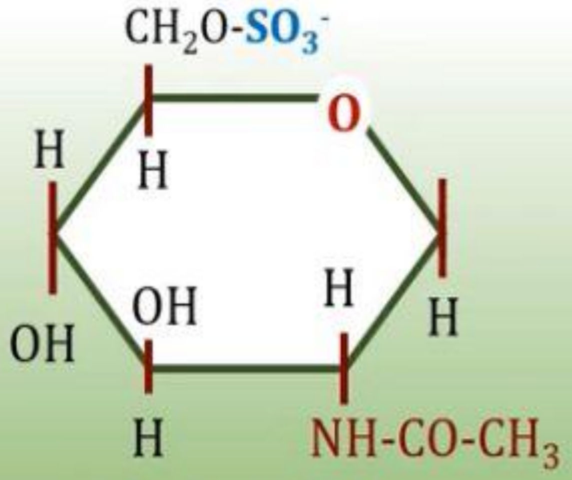
Galactosamine

Glycosaminoglycans

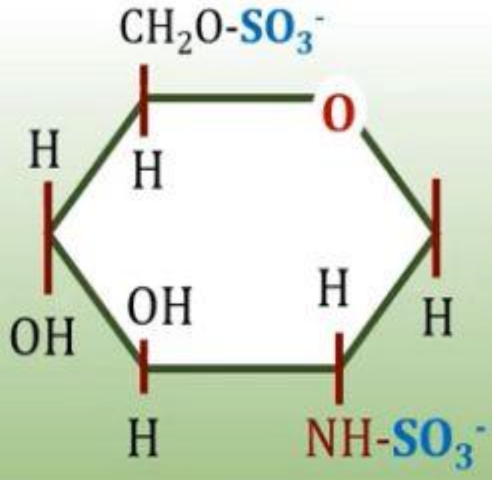
- Amino sugars
  - Glucosamine
  - Galactosamine
- Uronic acids



N-Acetylglucosamine



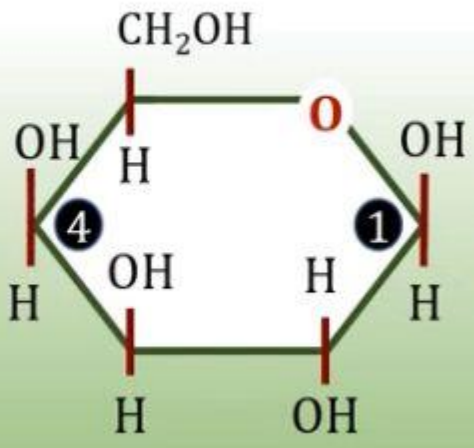
N-Acetylglucosamine 6-sulphate



N-Sulpho-glucosamine 6-sulphate

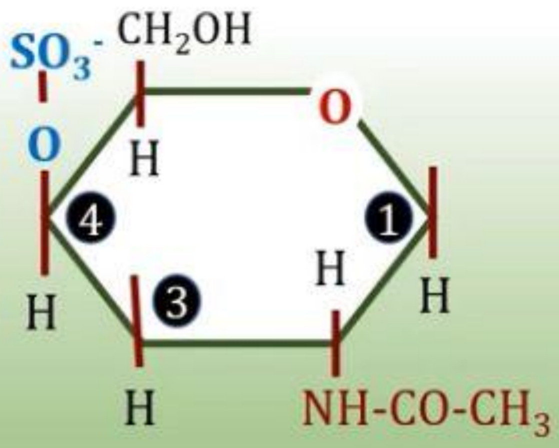
Glycosaminoglycans

- Amino sugars
  - Glucosamine
  - Galactosamine
- Uronic acids



β-D Galactose

Keratan sulphate



N-Acetylgalactosamine 4-sulphate

Chondroitin sulphate, Dermatan sulphate

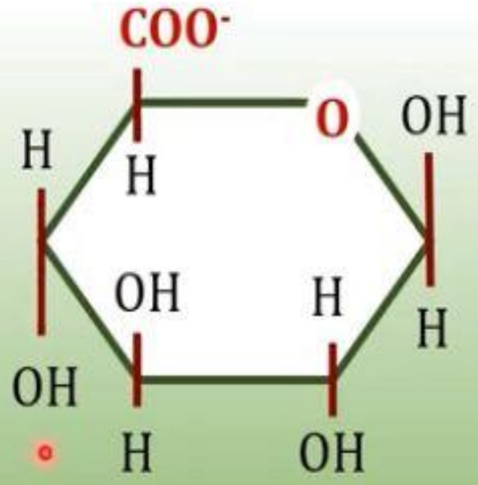
Glycosaminoglycans

Amino sugars

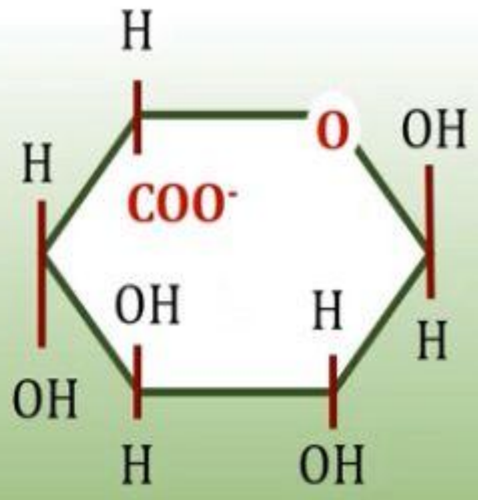
- Glucosamine
- Galactosamine

Uronic acids

- Glucuronic acid
- Iduronic acids

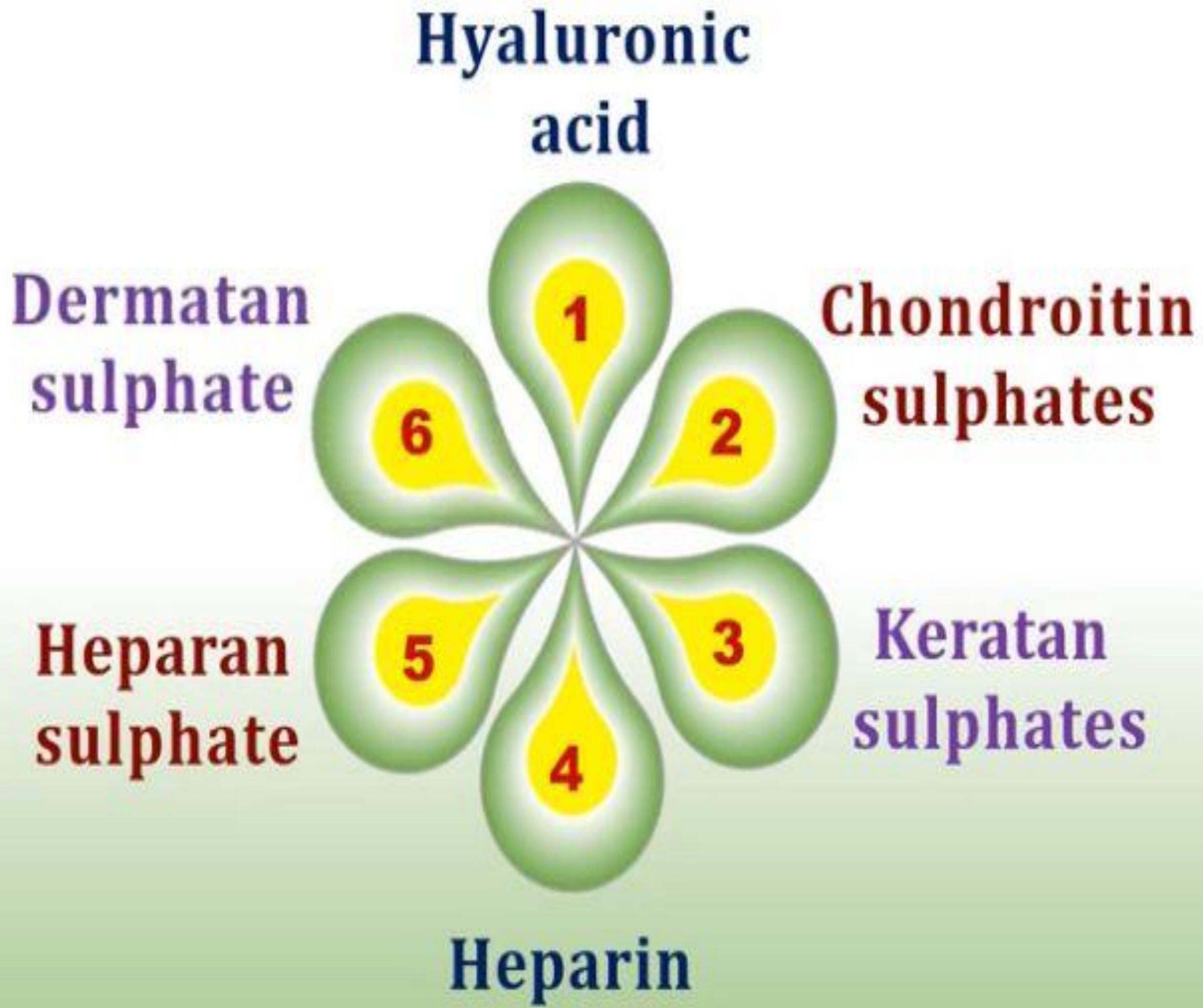


Glucuronic acid

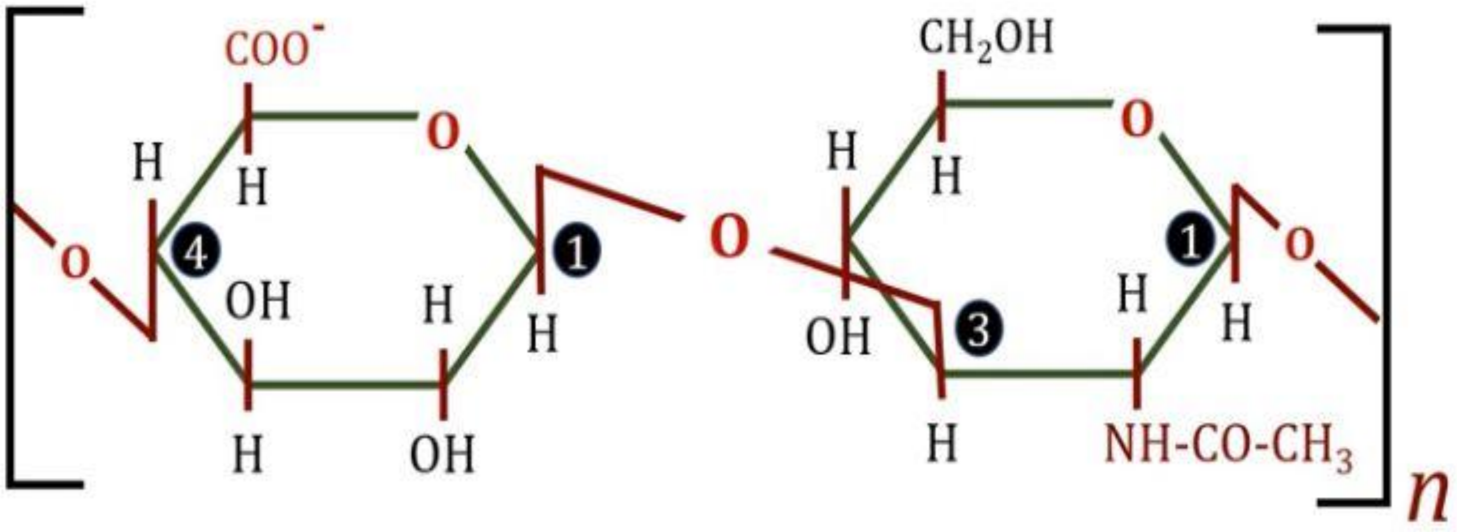


Iduronic acid

Glycosaminoglycans:Examples



1: Hyaluronic Acid



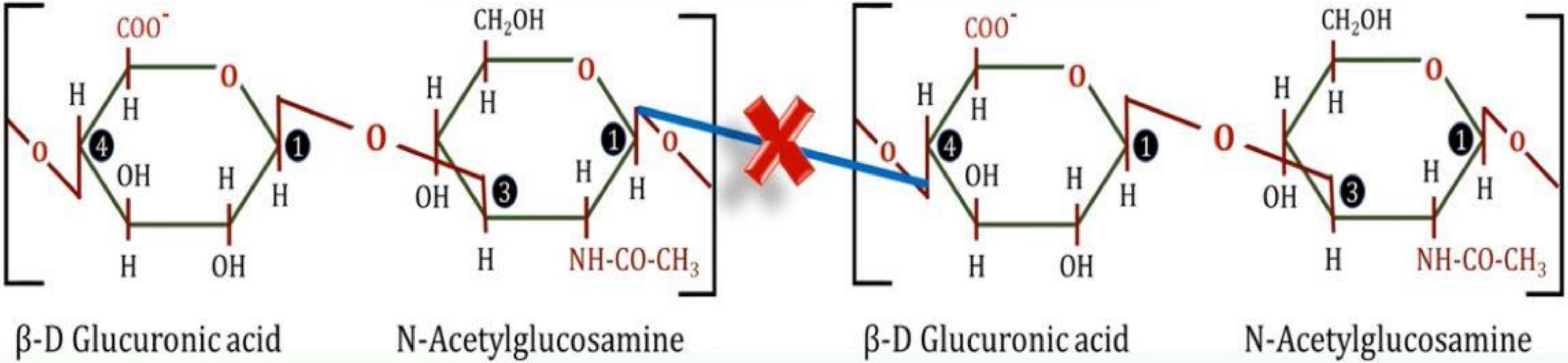
β-D Glucuronic acid N-Acetylglucosamine

β-(1-3)- glycosidic linkage

- Different from other GAGs: Unsulfated, not covalently attached to protein
- Only GAG not limited to animal tissue, but also found in bacteria.
- Found in synovial fluid of joints, vitreous humor of the eye, the umbilical cord, loose connective tissue and cartilage.

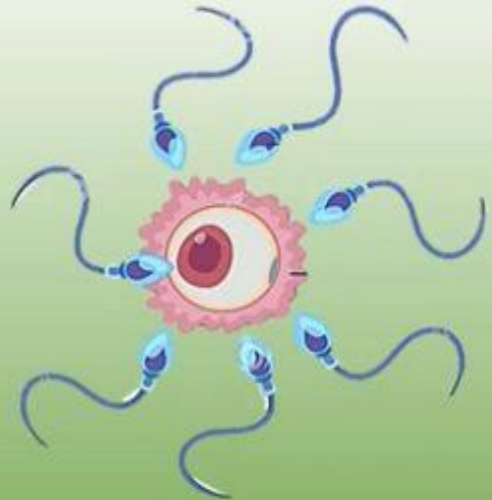
Hyaluronic Acid

Contain 250-25,000 disaccharide units

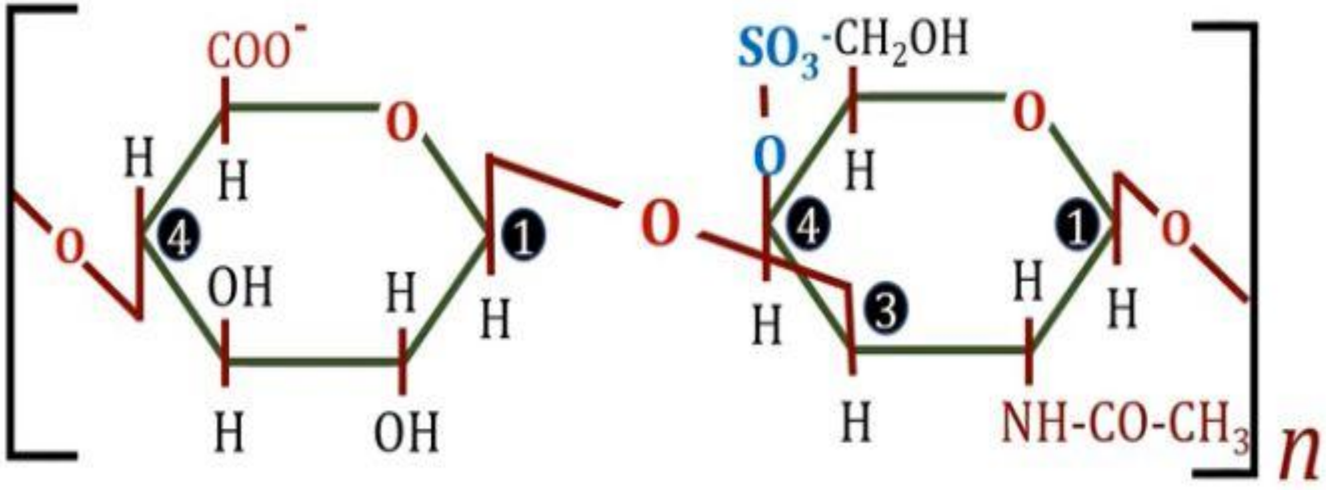


β-(1-4)- glycosidic linkage

- Hyaluronidase breaks **β-(1-4)- glycosidic linkage**
- Present in testes, seminal fluid, certain snake & insect venoms
- **Hyaluronidase of semen : Important role in fertilization**

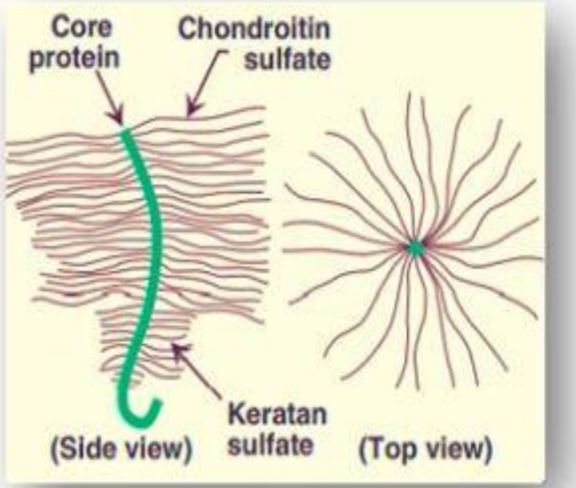


Chondroitin sulphate



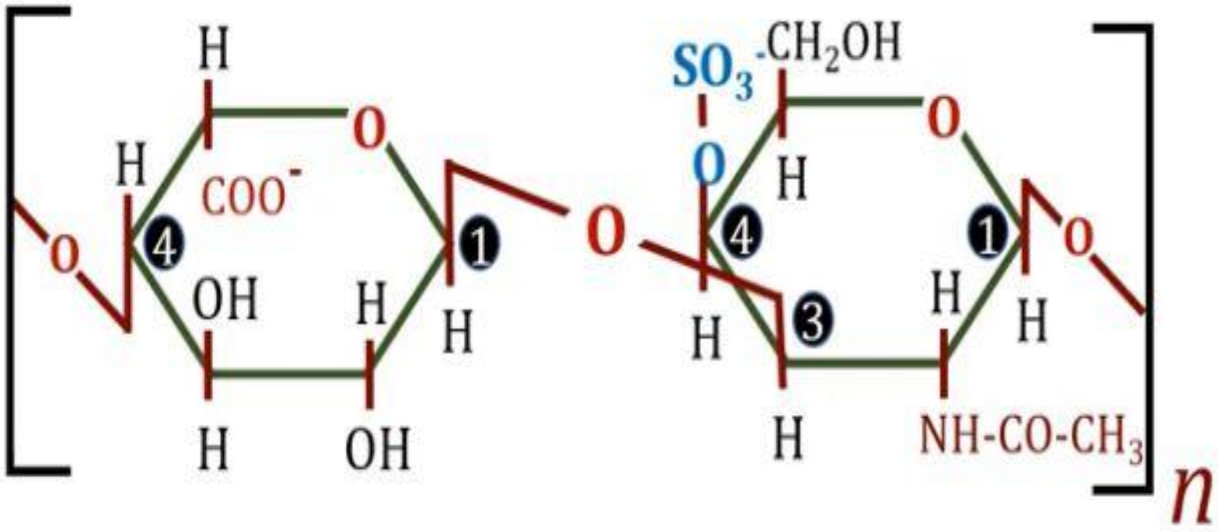
β-D Glucuronic acid N-Acetylgalactosamine 4-sulphate

β-(1-3)- glyosidic linkage



- Most abundant GAGs in the body.
- Found in cartilage, tendons, ligaments, and aorta.
- Form proteoglycan aggregates, often aggregating noncovalently with hyaluronic acid
- In cartilage, they bind collagen and hold fibers in a tight, strong network
- Helps to maintain the structure and shape of tissues.

Dermatan sulphate



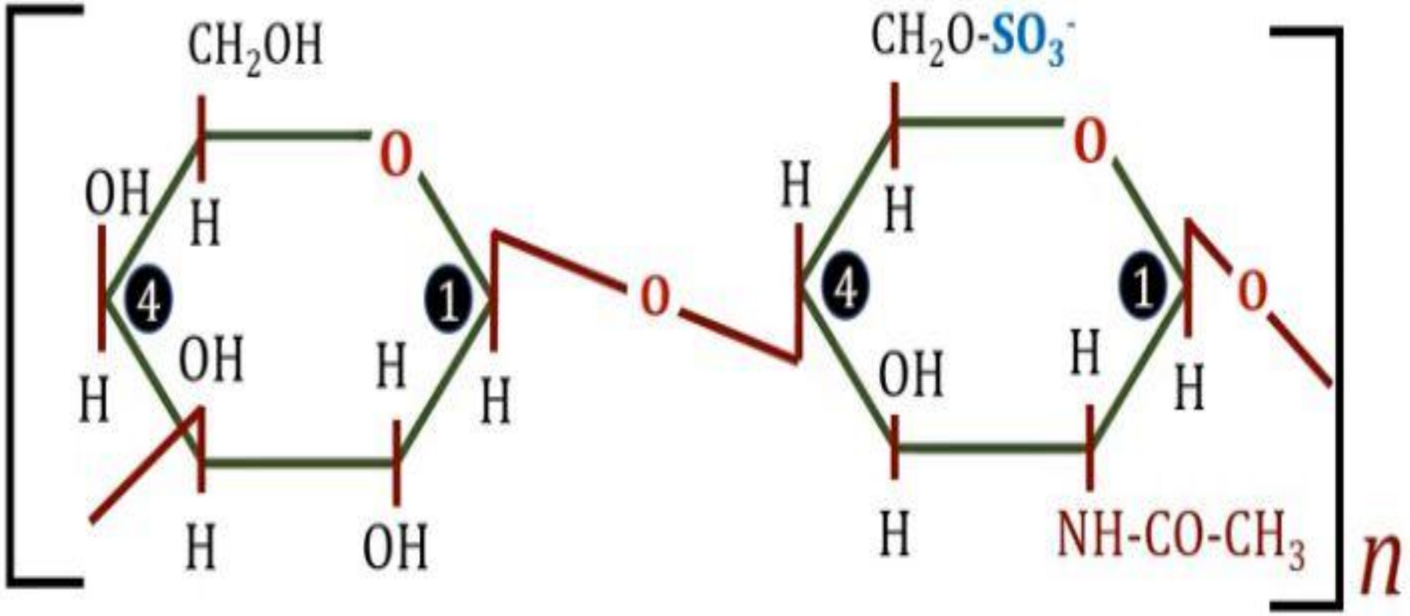
L-Iduronic acid N-Acetylgalactosamine 4-sulphate

(with variable amounts of glucuronic acid).

β-(1-3)- glycosidic linkage

- It is found in skin, blood vessels and heart valves
- Maintains the shape of skin.

Keratan Sulphate

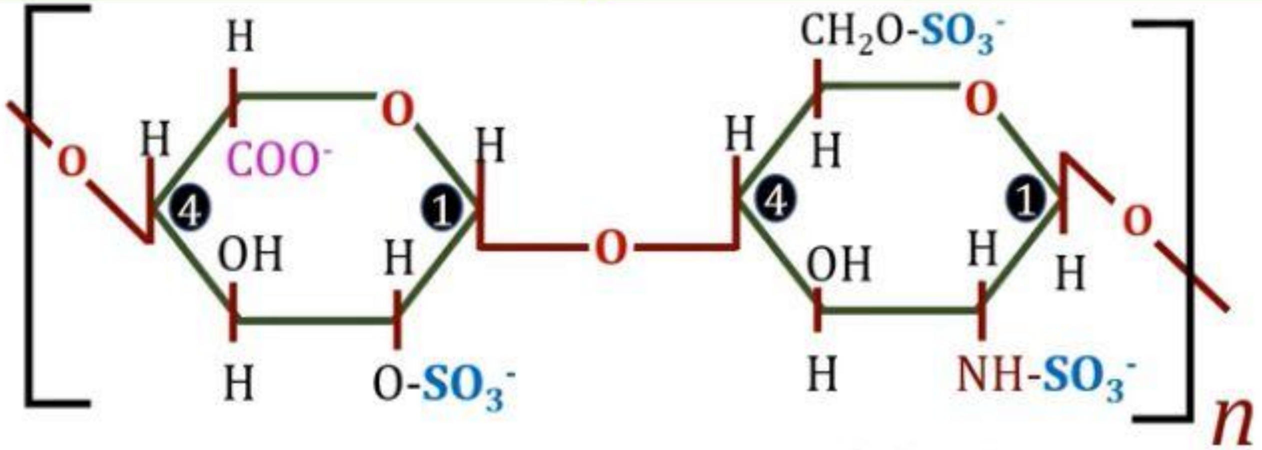


β-D Galactose N-Acetylglucosamine 6-sulphate

β-(1-4)- glycosidic linkage

- It is the only GAG which does not contain any uronic acid.
- It is found in cornea and tendons.
- Keeps cornea transparent

Heparin



L-Iduronic acid-2-sulphate (Or Glucoronic acid) N-Sulphoglucosamine 6-sulphate

α-(1-4)- glycosidic linkage

- Unlike other GAGs (extracellular), heparin is an intracellular component of mast cells that line arteries especially in liver, lungs, and skin.
- Serves as an anticoagulant.
- It activates antithrombin III, which in turn inactivates thrombin, so blood is not clotted.

- **Heparan sulfate**: Disaccharide unit: Same as heparin except some glucosamines are acetylated and there are fewer sulfate groups.
- Extracellular GAG, found in basement membrane and as a ubiquitous component of cell surfaces.

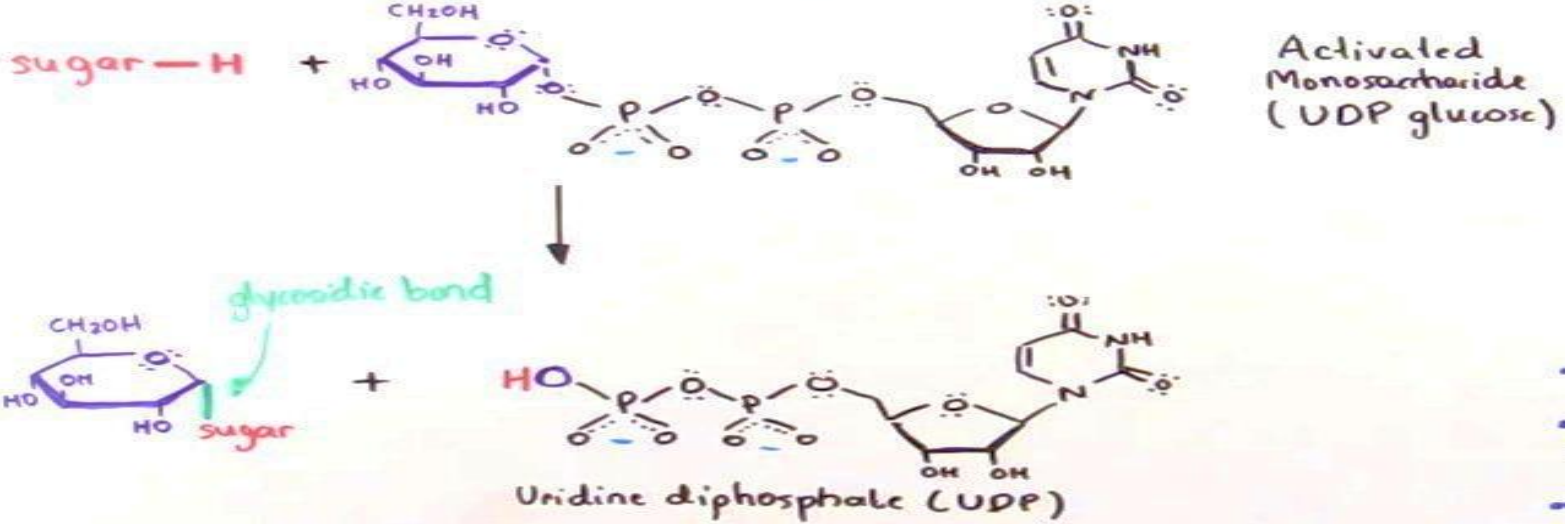
GAGs: Summary

Hyaluronic acid	Glucuronic acid, N-acetylglucosamine	Unsulfated Not covalently attached Also present in Bacteria
Heparin	L-Iduronic acid (Glucuronic acid), Sulfated glucosamine	Intracellular, anticoagulant
Keratan sulphate	**Galactose,** N-acetylglucosamine	No Uronic acid Corneal transparency
Chondroitin Sulphate	Glucuronic acid, **N-acetyl galactosamine**	Most abundant GAG
Dermatan sulphate	L-Iduronic acid, **N-acetyl galactosamine**	

Blood group substances (blood gr Antigens)

- RBC membrane contains several antigenic substance, based on which classified into different blood groups.
- They contain carbohydrates as glycoproteins or glycolipids.
- N-Acetylgalactosamine, galactose, fucose, sialic acid etc are found in blood gr substances.
- Carbohydrate content plays a determinant role in blood grouping.

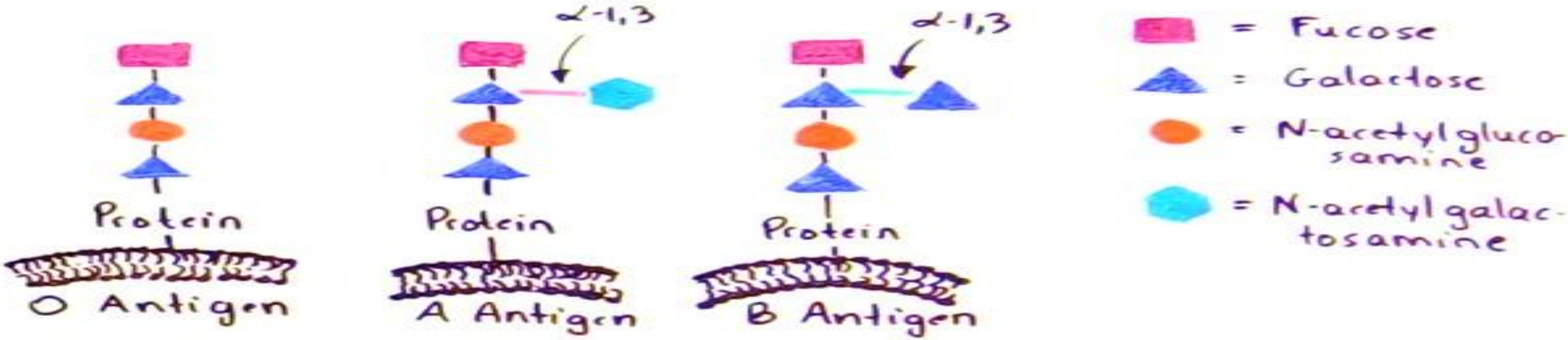
- Glycosyltransferases are enzymes that catalyze the formation of glycosidic bonds. This can mean joining individual monosaccharides to form larger polysaccharides or joining sugars to other biomolecules such as proteins.
- Glycosyltransferases are highly-specific molecules; each glycosyltransferase carries out a specific glycosidic linkage.



- Glycosyltransferase uses an activated sugar, usually a sugar nucleotide such as UDP glucose, to catalyze the formation of a glycosidic bond.

ABO Blood Type

- The ABO blood group is determined by the type of glycosyltransfrase that is expressed in the individual.
- The antigens in the ABO system are O-glycosylated glycoproteins that are bound onto red blood cell membranes. The terminal sugar residues on the antigen determines the group type.



- All three antigens have the O-oligosaccharide sequence in common.
- A specific glycosyltransferase must be present in order to add an extra sugar to the galactose of the O-oligosaccharide.
- A <u>type A glycosyltransferase</u> is needed to add the N-acetylgalatosamine while a <u>type B glycosyltransferase</u> is needed to add the galactose to the O-oligosaccharide.
- Blood type O individuals lack the genes that express both type A and type B glycosyltransferases.